

The *Anytime* Convergence of Stochastic Gradient Descent with Momentum: From a Continuous-Time Perspective

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Abstract

We study the stochastic optimization problem from a continuous-time perspective, with a focus on the Stochastic Gradient Descent with Momentum (SGDM) method. We show that the trajectory of SGDM, despite its *stochastic* nature, converges in L_2 -norm to a *deterministic* second-order Ordinary Differential Equation (ODE) as the stepsize goes to zero. The connection between the ODE and the algorithm results in a useful development for the discrete-time convergence analysis. More specifically, we develop, through the construction of a suitable Lyapunov function, convergence results for the ODE, which are then translated to the corresponding convergence results for the discrete-time case. This approach yields a novel *anytime* convergence guarantee for stochastic gradient methods. In particular, we prove that the sequence $\{x_k\}$, governed by running SGDM on a smooth convex function f , satisfies

$$\mathbb{P}\left(f(x_k) - f^* \leq C \left(1 + \log \frac{1}{\beta}\right) \frac{\log k}{\sqrt{k}}, \text{ for all } k\right) \geq 1 - \beta \quad \text{for any } \beta > 0,$$

where $f^* = \min_{x \in \mathbb{R}^n} f(x)$, and C is a constant. Rather than at a single step, this result captures the convergence behavior across the entire trajectory of the algorithm.

1 Introduction

Stochastic gradient methods are widely used in various fields such as machine learning (e.g., Johnson and Zhang, 2013; Loizou and Richtárik, 2017), statistics (e.g., Robbins and Monro, 1951), and more. These algorithms, including Stochastic Gradient Descent (SGD) and its variants, have been successfully implemented in many applications. Many research works have focused on establishing convergence guarantees for SGD (Nemirovski et al., 2009; Shamir and Zhang, 2013; Harvey et al., 2019) and its variants (Lan, 2012; Ghadimi and Lan, 2013; Gitman et al., 2019; Li and Orabona, 2019; Gorbunov et al., 2020; Liu et al., 2020), aiming to ensure their efficiency and reliability.

Also, the relationship between the continuous-time dynamics and the discrete-time recursive algorithms has been studied in mathematics and related fields (Ljung, 1977; Helmke and Moore, 1996). Recently, there is a resurgence of interest in the understanding of continuous-time interpretation of discrete-time algorithms, inspired by the ODE modeling of Nesterov’s accelerated gradient method (Su et al., 2014). This new perspective has led to valuable insights in the analysis (Wibisono et al., 2016; Shi et al., 2021) and design (Krichene et al., 2015; Zhang et al., 2018; Shi et al., 2019; Zhang et al., 2021) of the deterministic optimization algorithms.

The relationship between continuous-time dynamics and discrete-time algorithms has also been explored in the context of stochastic optimization (Mandt et al., 2016; Li et al., 2017; He et al., 2018; Orvieto

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and Lucchi, 2019; Cheng et al., 2020; Shi, 2021; Difonzo et al., 2022; Shi et al., 2023). In particular, previous work by Cheng et al. (2020) investigated numeric SDEs in machine learning applications, motivated by stochastic gradient algorithms. However, the continuous-time limits of some stochastic optimization algorithms, including the SGDM in this paper, exhibit different behavior. Specifically, the stochasticity may disappear as the stepsize goes to zero. Another line of research has centered around Langevin dynamics (e.g., Hsieh et al., 2018; Xu et al., 2018). With roots in the ergodic analysis of simulated annealing and related algorithms (Chiang et al., 1987; Gelfand and Mitter, 1991; Bélisle, 1992; Granville et al., 1994; Mattingly et al., 2010), recent studies in the machine learning community have primarily focused on specific tasks, such as statistical sampling for machine learning (Welling and Teh, 2011; Zou et al., 2019; Chatterji et al., 2020; Mou et al., 2021) and Riemannian learning (Cheng et al., 2022).

In this paper, we consider the optimization problem, $\min_{x \in \mathbb{R}^n} f(x)$, in an environment where f is accessible only through a stochastic gradient oracle. That is, if we query the oracle at x , it returns a stochastic version $g(x, \xi)$ of the gradient $\nabla f(x)$. We use x^* to denote an optimal solution of this problem, and $f^* = f(x^*) = \min_{x \in \mathbb{R}^n} f(x)$ is the minimal function value. Our main focus is on studying the Stochastic Gradient Descent with Momentum (SGDM) method given by

$$\begin{aligned} x_k - x_{k-1} &= \eta v_{k-1}, \\ v_k - v_{k-1} &= -\frac{2}{k} v_k - \frac{2}{k} \cdot \frac{1}{\sqrt{k\eta}} g(x_k, \xi_k), \end{aligned} \tag{1}$$

where η is the stepsize. From a continuous-time perspective, we rigorously prove that as η tends to zero, the sequence $\{x_k\}$ generated by SGDM converges in L_2 -norm to the trajectory of the second-order ODE

$$\ddot{X}(t) + \frac{2}{t} \dot{X}(t) + \frac{2}{t^{3/2}} \nabla f(X(t)) = 0. \tag{2}$$

The continuous-time limits of some other stochastic optimization methods are also ODEs of the form (2), including the well-known AC-SA algorithm (Lan, 2020); see Appendix A for a detailed derivation.

This continuous-time version provides insights into the discrete-time analysis. In this connection, we establish the convergence guarantees of the ODE (2), which are then translated into the convergence results of SGDM. Specifically, this approach gives a new characterization of the dynamical property of SGDM. Leveraging this dynamical insight, we develop a novel supermartingale argument and contribute to the existing theoretical analysis on stochastic gradient methods, particularly in terms of the anytime high probability convergence guarantee under mild conditions.

1.1 The Anytime High Probability Convergence and Our Contributions

For theoretical analysis of stochastic optimization algorithms, an important topic is to develop convergence guarantees that hold with high probability (Nemirovski et al., 2009; Lan, 2012; Nazin et al., 2019; Gorbunov et al., 2020; Liu et al., 2023). This is partly due to the fact that high probability analysis better reflects the performance of a single run of the algorithm. For a given probability $1 - \beta$, the existing results typically take the form:

$$\text{For each } k \in \mathbb{N}, \mathbb{P}(f(x_k) - f^* \leq U(k, \beta)) \geq 1 - \beta, \tag{3}$$

where $U(k, \beta)$ represents an upper bound depending on k and β . However, the guarantee (3) is insufficient if a practitioner wants to capture the behavior of the whole trajectory of the algorithm, or requires the upper bound to remain valid with high probability at any data-dependent stopping time (Waudby-Smith and Ramdas, 2020; Howard et al., 2021). For instance, multi-fidelity hyperparameter optimization (HPO) algorithms (Jamieson and Talwalkar, 2016; Li et al., 2018) use losses at various training iterates, rather than only at the last iterate, in the hyperparameter selection, and thus accelerate the HPO process. Therefore, the convergence bound of the whole training loss trajectory is important for the effectiveness of the HPO algorithm. In such cases, the following condition for the sequence $\{x_k\}$ must be met:

$$\mathbb{P}(f(x_k) - f^* \leq U(k, \beta), \text{ for all } k) \geq 1 - \beta. \tag{4}$$

We shall refer to results of the form (4) as *anytime high probability guarantees*. Furthermore, this concept is closely related to the *confidence sequence*, which has been widely investigated in sequential statistics (Lai, 1976; Sinha and Sarkar, 1984) and applied to online learning (Jamieson et al., 2014; Kaufmann and Koolen, 2021; Orabona and Jun, 2023). However, to the best of our knowledge, anytime convergence guarantees have not been studied in the stochastic optimization literature.

In this paper, we leverage a new continuous-time perspective to provide the anytime high probability result of the form (4) for stochastic gradient methods. Notably, our *anytime* convergence achieves the same rate as the state-of-the-art results for a *single* iterate (Lan, 2012; Liu et al., 2023). Our main contributions are as follows.

1. We are, to the best of our knowledge, the first to rigorously show that the continuous-time L_2 -norm limit of a momentumized stochastic gradient method is a deterministic second-order ODE. We are the first to rigorously handle the quadratic variation process in analyzing the continuous-time limit of momentumized stochastic gradient method. This quadratic variation term is crucial in Itô's calculus. We develop convergence results for this ODE through a Lyapunov function, and then use it as a guidance for the algorithm analysis.
2. We translate the whole continuous-time argument to the discrete-time case to construct the Lyapunov function of SGDM, $\mathcal{E}(k)$, and establish its descent property. This approach reveals the intrinsic dynamic property of SGDM. Furthermore, the increment of the Lyapunov function, $\mathcal{E}(k+1) - \mathcal{E}(k)$, is itself bounded by its previous value. As a consequence, our discrete-time analysis does not rely on any projection, or the assumption of bounded gradients or stochastic gradients.
3. We establish the first explicit *anytime* high probability convergence guarantee for stochastic gradient methods. We prove that the sequence $\{x_k\}$ governed by running SGDM on a smooth convex function f satisfies

$$\mathbb{P} \left(f(x_k) - f^* \leq \frac{C \log k \left(1 + \log \frac{1}{\beta}\right)}{\sqrt{k}}, \text{ for all } k \right) \geq 1 - \beta$$

for any β , where C is a constant. As far as we know, no prior research on stochastic gradient methods has studied the probability that the convergence bound holds simultaneously over iterates. In this work, we propose a novel procedure for proving this anytime high probability convergence without losing any factors in k , compared to its single-iterate counterpart. More specifically, based on the descent property of the Lyapunov function, we derive a Gronwall-type inequality. Then our key approach is to construct a supermartingale by altering the moment generating function of $\mathcal{E}(k)$, which leads to the anytime convergence bound.

4. For smooth convex objective functions, we show that with probability 1 there exists a sub-sequence $\{x_{k_l}\}$ of $\{x_k\}$ generated by SGDM satisfies $f(x_{k_l}) - f^* = o\left(\frac{1}{\sqrt{k_l} \log \log k_l}\right)$. This rate is new to the best of our knowledge.

1.2 Comparison to existing results

We illustrate the qualitative and quantitative improvements of our *anytime* high probability convergence results. The high probability guarantees for stochastic gradient methods were obtained in Lan (2012); Harvey et al. (2019); Liu et al. (2023); Liu and Zhou (2023). It has been shown that for a fix number of iterations k , with probability greater than $1 - \beta$,

$$\frac{1}{k} \sum_{l=1}^k f(x_l) - f^* \leq U(k, \beta) \quad \text{and} \quad f(x_k) - f^* \leq U(k, \beta),$$

where $U(k, \beta)^1$ is the asymptotic convergence rate given by

$$U(k, \beta) = C \left(1 + \log \frac{1}{\beta} \right) \frac{\log k}{\sqrt{k}}.$$

One may derive a naive anytime high probability guarantee by setting $\beta = \beta/k^2$ and taking the union bounds. This indeed leads to

$$\mathbb{P} \left(f(x_k) - f^* \leq \tilde{U}(k, \beta), \text{ for all } k \right) \geq 1 - \mathcal{O}(\beta),$$

with

$$\tilde{U}(k, \beta) = U(k, \beta) \log k.$$

Our anytime high probability convergence guarantee, however, shows that the extra $\log k$ factor can be reduced. The *anytime* convergence rate we obtained in Theorem 5 is (surprisingly) of the same order as the ones for the average iterate and the last iterate. Therefore, our *anytime* convergence guarantee achieves an $\mathcal{O}(\log k)$ improvement compared to the state-of-the-art result (Liu et al., 2023). We remark that it is due to the novel energy estimate in Lemma 3 and an application of the supermartingale inequality.

Another potential approach to derive *anytime* high probability convergence is utilizing the almost sure convergence results. Sebbouh et al. (2021) derived almost sure convergence for various stochastic optimization algorithms, and in particular, the last-iterate almost sure convergence of stochastic heavy-ball algorithm in Theorem 13. By taking $\eta_k = \log k / \sqrt{k}$, it has shown that, *almost surely*, $f(x_k) - f^* = o(\log k / \sqrt{k})$. In other words, for any β , there exists $C(\beta)$ such that

$$\mathbb{P} \left(f(x_k) - f^* \leq C(\beta) \frac{\log k}{\sqrt{k}}, \text{ for all } k \right) \geq 1 - \beta.$$

However, it is unclear how the constant $C(\beta)$ depends on β . Our results answer this question with an explicit $C(\beta) = C(1 + \log(1/\beta))$ under the sub-Gaussian assumption on the noise.

1.3 Existing High Probability Results

There have been substantial advancements in establishing high probability guarantees for stochastic gradient methods. Nemirovski et al. (2009) established convergence guarantees for a robust stochastic approximation approach. Lan (2012) proposed an accelerated method that is universally optimal for solving non-smooth, smooth, and stochastic problems. Ghadimi and Lan (2013) developed a two-phase method to improve the large deviation properties. Nazin et al. (2019), Madden et al. (2020), Gorbunov et al. (2020), Cutkosky and Mehta (2021), and Li and Liu (2022) proposed high probability convergence results without the light-tailed assumption. Li and Orabona (2020) and Kavis et al. (2021) studied stochastic gradient methods with adaptive stepsize. Liu et al. (2023) provided a new concentration argument to improve the high probability analysis. The last-iterate high probability convergence results for SGD is studied by Harvey et al. (2019). Jain et al. (2021) introduced a stepsize sequence to obtain the optimal last-iterate convergence bound. Liu and Zhou (2023) also presented a unified way for enhancing the last-iterate convergence analysis of stochastic gradient methods. Yet all of the existing works provide high probability results that hold for individual k , and none of them study the anytime convergence, or more specifically, the probability that the convergence guarantee holds simultaneously over all values of k . A survey of more related works on stochastic optimization is provided in Appendix B, including existing in expectation convergence results.

This paper is organized as follows. In Section 2, we prove that the continuous-time limit of SGDM is the ODE (2), and then construct a Lyapunov function to give the convergence result of (2). In Section 3, we translate the Lyapunov argument to the discrete case, which yields the discrete-time dynamical property. We then prove the convergence of SGDM in expectation. In addition, we demonstrate the almost sure convergence by the supermartingale convergence theorem. Moving on to Section 4, we utilize the

¹The bounds here are adapted to the setting of the current paper, where the learning rate is independent of β , see (Liu and Zhou, 2023, Theorem C.2) for full details.

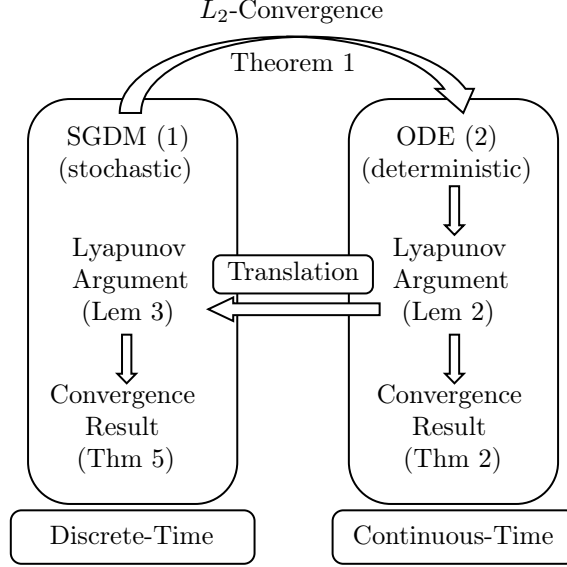


Figure 1: The framework of our theoretical analysis.

above results and establish the anytime high probability convergence guarantee of SGDM. We present the experimental results in Section 5, which confirm our theoretical findings. Figure 1 illustrates the entire framework of our theoretical analysis.

2 Continuous-Time Analysis

In this section, we show that the continuous-time limit of SGDM is the ODE (2), and construct a Lyapunov function to derive the convergence result of (2). Our analysis relies on the following assumption on the objective function f .

Assumption 1. f is convex and L -smooth, that is, $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ for any $x, y \in \mathbb{R}^n$. There exists $x^* = \arg \min_{x \in \mathbb{R}^n} f(x)$.

2.1 Derivation of the ODE Model

We rewrite (2) as the following system

$$\begin{cases} \frac{dX}{dt} = V, \\ \frac{dV}{dt} = -\frac{2}{t}V - \frac{2}{t^{\frac{3}{2}}}\nabla f(X). \end{cases} \quad (5)$$

Theorem 1 shows that the stochastic sequence $\{x_k\}$ generated by SGDM converges to the trajectory of the deterministic ODE system (5) as the stepsize goes to zero. For our purpose, we consider the ODE system (5) starting at an arbitrary $T_0 > 0$.

Theorem 1. Assume that f is L -smooth. Let $(X(t), V(t))$ be the solution to the ODE system (5), and $\{(x_k, v_k)\}$ be generated from SGDM (1) with $g(x_k, \xi_k) = \nabla f(x_k) + \xi_k$, where $\xi_k \stackrel{iid}{\sim} \mathcal{N}(0, I_n)$. If $X(T_0) = x_{T_0/\eta}$ and $V(T_0) = v_{T_0/\eta}$ for $T_0 > 0$, then

$$\lim_{\eta \rightarrow 0} \mathbb{E} [\|x_{T/\eta} - X(T)\|^2] = 0$$

holds for any $T > T_0$. Note that throughout we shell use the convention that T_0/η and T/η in subscripts of x and v are understood as their integer part.

The proof of Theorem 1 relies on constructing auxiliary SDEs with diminishing stochastic term, and stochastic calculus techniques calibrated to our setting. Before presenting the detailed proof, we shall give an informal derivation of the ODE (2) and (5).

We rewrite SGDM scheme (1) as

$$\begin{cases} x_k - x_{k-1} = \eta v_{k-1}, \\ v_k - v_{k-1} = -\frac{2}{k\eta} v_k \eta - \frac{2}{(k\eta)^{\frac{3}{2}}} \nabla f(x_k) \eta - \frac{2}{(k\eta)^{\frac{3}{2}}} (\sqrt{\eta} \xi_k) \sqrt{\eta}. \end{cases} \quad (6)$$

Let W be a standard Brownian motion. Then we let η in (6) go to zero and replace $x_k - x_{k-1}$, $v_k - v_{k-1}$, $k\eta$, η and $\sqrt{\eta} \xi$ by dX , dV , t , dt and dW , respectively. The discrete scheme (6) becomes

$$\begin{cases} dX = V dt, \\ dV = -\frac{2}{t} V dt - \frac{2}{t^{\frac{3}{2}}} \nabla f(X) dt - \frac{2}{t^{\frac{3}{2}}} \sqrt{dt} dW. \end{cases}$$

Since $\sqrt{dt} dW$ is a higher order term, we omit it and obtain (5), which leads to the ODE (2) by subscribing the first line to the second line.

Proof of Theorem 1. For any fixed $\eta > 0$, we write $t_k = k\eta$ and consider an auxiliary SDE system $(\tilde{X}(t), \tilde{V}(t))$ on $[T_0, T]$ with initialization $\tilde{X}(T_0) = x_{T_0/\eta}$ and $\tilde{V}(T_0) = v_{T_0/\eta}$, which is defined as

$$\begin{cases} d\tilde{X}(t) = \tilde{V}(t_k) dt, \\ d\tilde{V}(t) = -\frac{2}{t_k} \tilde{V}(t_k) dt - \frac{2}{t_k^{\frac{3}{2}}} \nabla f(\tilde{X}(t_k)) dt - \frac{2}{t_k^{\frac{3}{2}}} \sqrt{\eta} dW(t), \end{cases} \quad (7)$$

for $t \in [t_k, t_{k+1})$. Integrating sequentially on each interval $[t_k, t_{k+1})$ shows that $(\tilde{X}(t_k), \tilde{V}(t_k))$ and (x_k, v_k) are identically distributed for each $T_0 \leq t_k \leq T$. For convenience, we denote

$$\tilde{Y}(t) = \begin{pmatrix} \tilde{X}(t) \\ \tilde{V}(t) \end{pmatrix} \quad \text{and} \quad Y(t) = \begin{pmatrix} X(t) \\ V(t) \end{pmatrix},$$

where $X(t)$ and $V(t)$ are defined in the ODE system (5), and thus $\mathbb{E}[\|x_{T/\eta} - X(T)\|^2] = \mathbb{E}[\|\tilde{X}(T) - X(T)\|^2] \leq \mathbb{E}[\|\tilde{Y}(T) - Y(T)\|^2]$. The following lemma shows that $\{\tilde{Y}(t_k)\}$ are uniformly bounded in L_2 -norm.

Lemma 1. *Let $[T_0, T]$ be a fixed horizon. There exists a constant C_1 such that for any $\eta > 0$,*

$$\max_{\frac{T_0}{\eta} \leq k \leq \frac{T}{\eta}} \mathbb{E}[\|\tilde{Y}(t_k)\|^2] \leq C_1, \quad (8)$$

where $t_k = k\eta$. Furthermore, there exists a constant C_2 such that for any $\eta > 0$ and any $T_0 \leq t_k \leq t \leq t_{k+1} \leq T$,

$$\mathbb{E}[\|\tilde{Y}(t) - \tilde{Y}(t_k)\|^2] \leq C_2 \eta^2. \quad (9)$$

Proof. From (7), it follows that

$$\tilde{Y}(t) - \tilde{Y}(t_k) = \begin{pmatrix} \tilde{V}(t_k) \\ -\frac{2}{t_k} \tilde{V}(t_k) - \frac{2}{t_k^{\frac{3}{2}}} \nabla f(\tilde{X}(t_k)) \end{pmatrix} \cdot (t - t_k) + \begin{pmatrix} 0 \\ -\frac{2}{t_k^{\frac{3}{2}}} \sqrt{\eta} (W(t) - W(t_k)) \end{pmatrix},$$

for any $t_k \leq t \leq t_{k+1}$. By the L -smoothness,

$$\|\nabla f(x)\| = \|\nabla f(x) - \nabla f(x^*)\| \leq L \cdot \|x - x^*\| \leq L\|x\| + L\|x^*\|. \quad (10)$$

Therefore, we have

$$\begin{aligned} \|\tilde{Y}(t) - \tilde{Y}(t_k)\|^2 &\leq 2 \left(\|\tilde{V}(t_k)\|^2 + \frac{8}{t_k^2} \|\tilde{V}(t_k)\|^2 + \frac{16L^2}{t_k^3} \left(\|\tilde{X}(t_k)\|^2 + \|x^*\|^2 \right) \right) (t - t_k)^2 \\ &\quad + \frac{8}{t_k^3} \eta (W(t) - W(t_k))^2, \end{aligned}$$

and thus there exists a constant C_3 , depending on T_0 , $\|x^*\|$ and L , such that

$$\|\tilde{Y}(t) - \tilde{Y}(t_k)\|^2 \leq C_3 \left(\|\tilde{Y}(t_k)\|^2 + 1 \right) (t - t_k)^2 + C_3 \eta (W(t) - W(t_k))^2.$$

Since $t \leq t_{k+1} = t_k + \eta$, taking expectation on both sides yields

$$\mathbb{E} \left[\|\tilde{Y}(t) - \tilde{Y}(t_k)\|^2 \right] \leq 2C_3 \eta^2 \left(\mathbb{E} \left[\|\tilde{Y}(t_k)\|^2 \right] + 1 \right). \quad (11)$$

Specifically, by letting $t = t_{k+1}$ we have

$$\mathbb{E} \left[\|\tilde{Y}(t_{k+1}) - \tilde{Y}(t_k)\|^2 \right] \leq 2C_3 \eta^2 \left(\mathbb{E} \left[\|\tilde{Y}(t_k)\|^2 \right] + 1 \right).$$

We denote

$$a_n = \max_{\frac{T_0}{\eta} \leq k \leq \frac{T_0}{\eta} + n} \left(\mathbb{E} \left[\|\tilde{Y}(t_k)\|^2 \right] \right)^{\frac{1}{2}},$$

and we have

$$\begin{aligned} a_{n+1} &\leq a_n + \left(\mathbb{E} \left[\|\tilde{Y}(t_{k+1}) - \tilde{Y}(t_k)\|^2 \right] \right)^{\frac{1}{2}} \leq a_n + \sqrt{2C_3} \eta \left(\mathbb{E} \left[\|\tilde{Y}(t_k)\|^2 \right] + 1 \right)^{\frac{1}{2}} \\ &\leq (1 + \sqrt{2C_3} \eta) a_n + \sqrt{2C_3} \eta. \end{aligned}$$

Therefore, for $N = \frac{T-T_0}{\eta}$, we have

$$\begin{aligned} a_N &\leq (1 + \sqrt{2C_3} \eta)^N a_0 + \sqrt{2C_3} \eta \sum_{k=0}^{N-1} (1 + \sqrt{2C_3} \eta)^k \\ &\leq \left(1 + \frac{\sqrt{2C_3}(T-T_0)}{N} \right)^N a_0 + \frac{\sqrt{2C_3}(T-T_0)}{N} \cdot N \left(1 + \frac{\sqrt{2C_3}(T-T_0)}{N} \right)^N \\ &\leq e^{\sqrt{2C_3}(T-T_0)} \left(\mathbb{E} \left[\|\tilde{Y}(T_0)\|^2 \right] \right)^{\frac{1}{2}} + \sqrt{2C_3}(T-T_0) e^{\sqrt{2C_3}(T-T_0)}. \end{aligned}$$

Thus, we finish the proof of (8) by letting

$$C_1 = \left(e^{\sqrt{2C_3}(T-T_0)} \left(\mathbb{E} \left[\|\tilde{Y}(T_0)\|^2 \right] \right)^{\frac{1}{2}} + \sqrt{2C_3}(T-T_0) e^{\sqrt{2C_3}(T-T_0)} \right)^2.$$

For the second part, we apply (8) to the RHS of (11) to get

$$\mathbb{E} \left[\|\tilde{Y}(t) - \tilde{Y}(t_k)\|^2 \right] \leq 2C_3 \eta^2 (C_1 + 1) = C_2 \eta^2,$$

where $C_2 = 2C_3(C_1 + 1)$. □

In the remaining proof, we will upper bound

$$d\mathbb{E}\left[\|\tilde{Y}(t) - Y(t)\|^2\right] = d\mathbb{E}\left[\|\tilde{X}(t) - X(t)\|^2\right] + d\mathbb{E}\left[\|\tilde{V}(t) - V(t)\|^2\right], \quad (12)$$

and then apply Grönwall's inequality to $\mathbb{E}\left[\|\tilde{Y}(t) - Y(t)\|^2\right]$ to get the convergence result of Theorem 1.

For the first term on the RHS of (12), the Itô's formula yields that

$$d\|\tilde{X}(t) - X(t)\|^2 = 2(\tilde{X}(t) - X(t))^\top d(\tilde{X}(t) - X(t)) + d\text{tr}[\tilde{X} - X](t),$$

where $[\tilde{X} - X]$ is the quadratic variation process of $\tilde{X} - X$. Since, by definitions (5) and (7), $d(\tilde{X}(t) - X(t)) = (\tilde{V}(t_k) - V(t))dt$, $\tilde{X} - X$ is absolutely continuous with respect to dt . Therefore, its quadratic variation must be zero, i.e., $[\tilde{X} - X](t) = 0$. We thus derive

$$\begin{aligned} & d\|\tilde{X}(t) - X(t)\|^2 \\ &= 2(\tilde{X}(t) - X(t))^\top (\tilde{V}(t_k) - \tilde{V}(t))dt + 2(\tilde{X}(t) - X(t))^\top (\tilde{V}(t) - V(t))dt \\ &\leq \|\tilde{X}(t) - X(t)\|^2 dt + \|\tilde{V}(t_k) - \tilde{V}(t)\|^2 dt + \|\tilde{X}(t) - X(t)\|^2 dt + \|\tilde{V}(t) - V(t)\|^2 dt. \end{aligned} \quad (13)$$

From (9) we have $\mathbb{E}\left[\|\tilde{V}(t_k) - \tilde{V}(t)\|^2\right] \leq C_2\eta^2$. Therefore, taking expectation on both sides of (13) gives

$$d\mathbb{E}\left[\|\tilde{X}(t) - X(t)\|^2\right] \leq 3\mathbb{E}\left[\|\tilde{Y}(t) - Y(t)\|^2\right] dt + C_2\eta^2 dt. \quad (14)$$

For the second term on the RHS of (12), the Itô's formula yields that

$$d\|\tilde{V}(t) - V(t)\|^2 = 2(\tilde{V}(t) - V(t))^\top d(\tilde{V}(t) - V(t)) + d\text{tr}[\tilde{V} - V](t). \quad (15)$$

The definitions (5) and (7) yields that

$$\begin{aligned} d(\tilde{V}(t) - V(t)) &= -\left(\frac{2}{t_k}\tilde{V}(t_k) - \frac{2}{t}V(t)\right)dt - \left(\frac{2}{t_k^{\frac{3}{2}}}\nabla f(\tilde{X}(t_k)) - \frac{2}{t^{\frac{3}{2}}}\nabla f(X(t))\right)dt \\ &\quad - \frac{2}{t_k^{\frac{3}{2}}}\sqrt{\eta}dW(t). \end{aligned}$$

Then the definition of quadratic variation gives

$$d\text{tr}[\tilde{V} - V](t) = \frac{4n}{t_k^3}\eta dt \leq \frac{4n}{T_0^3}\eta dt. \quad (16)$$

We also have

$$\begin{aligned} 2(\tilde{V}(t) - V(t))^\top d(\tilde{V}(t) - V(t)) &= -\underbrace{(\tilde{V}(t) - V(t))^\top \left(\frac{4}{t_k}\tilde{V}(t_k) - \frac{4}{t}V(t)\right)dt}_{\textcircled{1}} \\ &\quad - \underbrace{(\tilde{V}(t) - V(t))^\top \left(\frac{4}{t_k^{\frac{3}{2}}}\nabla f(\tilde{X}(t_k)) - \frac{4}{t^{\frac{3}{2}}}\nabla f(X(t))\right)dt}_{\textcircled{2}} \\ &\quad - \underbrace{(\tilde{V}(t) - V(t))^\top \cdot \frac{4}{t_k^{\frac{3}{2}}}\sqrt{\eta}dW(t)}_{\textcircled{3}}. \end{aligned} \quad (17)$$

The first part, ①, equals

$$\begin{aligned}
& - \left(\tilde{V}(t) - V(t) \right)^\top \left(\frac{4}{t_k} \tilde{V}(t_k) - \frac{4}{t} \tilde{V}(t) + \frac{4}{t} \tilde{V}(t) - \frac{4}{t} V(t) \right) dt \\
& = - \left(\tilde{V}(t) - V(t) \right)^\top \left(\frac{4}{t_k} \tilde{V}(t_k) - \frac{4}{t} \tilde{V}(t) \right) dt - \left(\tilde{V}(t) - V(t) \right)^\top \left(\frac{4}{t} \tilde{V}(t) - \frac{4}{t} V(t) \right) dt \\
& \leq \left\| \tilde{V}(t) - V(t) \right\|^2 dt + \left\| \frac{2}{t_k} \tilde{V}(t_k) - \frac{2}{t} \tilde{V}(t) \right\|^2 dt + \left\| \tilde{V}(t) - V(t) \right\|^2 dt + \frac{4}{T_0^2} \left\| \tilde{V}(t) - V(t) \right\|^2 dt.
\end{aligned} \tag{18}$$

We note that

$$\begin{aligned}
\left\| \frac{2}{t_k} \tilde{V}(t_k) - \frac{2}{t} \tilde{V}(t) \right\|^2 dt &= \left\| \frac{2}{t_k} \tilde{V}(t_k) - \frac{2}{t} \tilde{V}(t_k) + \frac{2}{t} \tilde{V}(t_k) - \frac{2}{t} \tilde{V}(t) \right\|^2 dt \\
&\leq 2 \left(\frac{2\eta}{T_0^2} \right)^2 \left\| \tilde{V}(t_k) \right\|^2 dt + \frac{4}{T_0^2} \left\| \tilde{V}(t_k) - \tilde{V}(t) \right\|^2 dt.
\end{aligned}$$

Lemma 1 gives that $\mathbb{E} \left[\left\| \tilde{V}(t_k) \right\|^2 \right] \leq C_1$ and $\mathbb{E} \left[\left\| \tilde{V}(t_k) - \tilde{V}(t) \right\|^2 \right] \leq C_2 \eta^2$. So, by taking expectation on both sides of the above inequality, we further have

$$\mathbb{E} \left[\left\| \frac{2}{t_k} \tilde{V}(t_k) - \frac{2}{t} \tilde{V}(t) \right\|^2 \right] dt \leq \left(\frac{8C_1}{T_0^4} + \frac{4C_2}{T_0^2} \right) \eta^2 dt. \tag{19}$$

Therefore, we denote $C_4 = \frac{8C_1}{T_0^4} + \frac{4C_2}{T_0^2}$, and (18) gives

$$\mathbb{E}[\textcircled{1}] \leq \left(2 + \frac{4}{T_0^2} \right) \mathbb{E} \left[\left\| \tilde{Y}(t) - Y(t) \right\|^2 \right] dt + C_4 \eta^2 dt. \tag{20}$$

For ②, we have

$$\begin{aligned}
\textcircled{2} &= - \left(\tilde{V}(t) - V(t) \right)^\top \left(\frac{4}{t_k^{\frac{3}{2}}} \nabla f(\tilde{X}(t_k)) - \frac{4}{t^{\frac{3}{2}}} \nabla f(\tilde{X}(t)) \right) dt \\
&\quad - \left(\tilde{V}(t) - V(t) \right)^\top \left(\frac{4}{t^{\frac{3}{2}}} \nabla f(\tilde{X}(t)) - \frac{4}{t^{\frac{3}{2}}} \nabla f(X(t)) \right) dt \\
&\leq \left\| \tilde{V}(t) - V(t) \right\|^2 dt + \left\| \frac{2}{t_k^{\frac{3}{2}}} \nabla f(\tilde{X}(t_k)) - \frac{2}{t^{\frac{3}{2}}} \nabla f(\tilde{X}(t)) \right\|^2 dt \\
&\quad + \left\| \tilde{V}(t) - V(t) \right\|^2 dt + \frac{4L^2}{T_0^3} \left\| \tilde{X}(t) - X(t) \right\|^2 dt.
\end{aligned} \tag{21}$$

Analogous to (19) and (10), there exists constant C_5 such that

$$\mathbb{E} \left[\left\| \frac{2}{t_k^{\frac{3}{2}}} \nabla f(\tilde{X}(t_k)) - \frac{2}{t^{\frac{3}{2}}} \nabla f(\tilde{X}(t)) \right\|^2 \right] dt \leq C_5 \eta^2 dt.$$

Therefore, (21) gives

$$\mathbb{E}[\textcircled{2}] \leq \left(2 + \frac{4L}{T_0^3} \right) \mathbb{E} \left[\left\| \tilde{Y}(t) - Y(t) \right\|^2 \right] dt + C_5 \eta^2 dt. \tag{22}$$

Additionally, by Lemma 1 we show ③ is a true martingale, and thus we derive

$$\mathbb{E}[\textcircled{3}] = 0. \tag{23}$$

As a consequence, taking expectation on both sides of (15) and substituting (16), (17), (20), (22), and (23) give

$$d\mathbb{E}\left[\|\tilde{V}(t) - V(t)\|^2\right] \leq \left(4 + \frac{4}{T_0^2} + \frac{4L}{T_0^3}\right) \mathbb{E}\left[\|\tilde{Y}(t) - Y(t)\|^2\right] dt + \left(C_4\eta^2 + C_5\eta^2 + \frac{4n}{T_0^3}\eta\right) dt. \quad (24)$$

Substituting (14) and (24) to (12) yields that

$$\begin{aligned} d\mathbb{E}\left[\|\tilde{Y}(t) - Y(t)\|^2\right] &\leq \left(7 + \frac{4}{T_0^2} + \frac{4L}{T_0^3}\right) \mathbb{E}\left[\|\tilde{Y}(t) - Y(t)\|^2\right] dt \\ &\quad + \left(C_2\eta^2 + C_4\eta^2 + C_5\eta^2 + \frac{4n}{T_0^3}\eta\right) dt. \end{aligned}$$

Without loss of generality, we let $\eta < 1$. Thus, there exist constants C_6 and C_7 such that

$$d\mathbb{E}\left[\|\tilde{Y}(t) - Y(t)\|^2\right] \leq C_6\mathbb{E}\left[\|\tilde{Y}(t) - Y(t)\|^2\right] dt + C_7\eta dt.$$

The Grönwall's inequality gives

$$\mathbb{E}\left[\|\tilde{Y}(T) - Y(T)\|^2\right] - \mathbb{E}\left[\|\tilde{Y}(T_0) - Y(T_0)\|^2\right] \leq \frac{C_7}{C_6} \left(e^{C_6(T-T_0)} - 1\right) \eta.$$

Since $\tilde{Y}(T_0) = Y(T_0)$ and $\mathbb{E}\left[\|x_{T/\eta} - X(T)\|^2\right] = \mathbb{E}\left[\|\tilde{X}(T) - X(T)\|^2\right] \leq \mathbb{E}\left[\|\tilde{Y}(T) - Y(T)\|^2\right]$, we complete the proof. $\square$

2.2 Convergence Results of the ODE

Here we consider a generalized version of ODE (2)

$$\ddot{X}(t) + \frac{p+1}{t} \dot{X}(t) + \frac{p+1}{t^\alpha} \nabla f(X(t)) = 0, \quad (25)$$

where α and p are parameters. Parameter α models the speed of the gradient-decaying. When setting $\alpha = \frac{3}{2}$ and $p = 1$, we recover ODE (2).

To describe the convergence of $X(t)$, we define the Lyapunov function

$$\mathcal{E}(t) = \left\|pX + t\dot{X} - px^*\right\|^2 + 2(p+1)t^{2-\alpha}(f(X) - f^*). \quad (26)$$

The following lemma shows that $\mathcal{E}(t)$ decreases monotonically with respect to t .

Lemma 2. *Let Assumption 1 hold, and X be the solution to (25) with $\alpha \leq 2$ and $p + \alpha \geq 2$. Then the Lyapunov function (26) satisfies*

$$\frac{d\mathcal{E}(t)}{dt} \leq 0.$$

Proof. By definition (26),

$$\begin{aligned} \frac{d\mathcal{E}(t)}{dt} &= 2 \left\langle pX + t\dot{X} - px^*, (p+1)\dot{X} + t\ddot{X} \right\rangle + 2(p+1)t^{2-\alpha} \left\langle \dot{X}, \nabla f(X) \right\rangle \\ &\quad + (4-2\alpha)(p+1)t^{1-\alpha}(f(X) - f^*). \end{aligned} \quad (27)$$

Since by (25), $(p+1)\dot{X} + t\ddot{X} = -(p+1)t^{1-\alpha}\nabla f(X)$, the above equality leads to

$$\begin{aligned} \frac{d\mathcal{E}(t)}{dt} &= -2p(p+1)t^{1-\alpha} \langle X - x^*, \nabla f(X) \rangle + (4-2\alpha)(p+1)t^{1-\alpha}(f(X) - f^*) \\ &= 2(p+1)t^{1-\alpha} \left((2-\alpha)(f(X) - f^*) - p \langle X - x^*, \nabla f(X) \rangle \right). \end{aligned}$$

From convexity of f , we have $f^* \geq f(X) + \langle \nabla f(X), x^* - X \rangle$. This inequality indicates that $\langle \nabla f(X), x^* - X \rangle \leq 0$. Then the condition $\alpha \leq 2$ and $p + \alpha \geq 2$ yields that

$$(2 - \alpha)(f(X) - f^*) - p \langle X - x^*, \nabla f(X) \rangle \leq (2 - \alpha)(f(X) - f^* - \langle X - x^*, \nabla f(X) \rangle) \leq 0,$$

and thus $\frac{d\mathcal{E}(t)}{dt} \leq 0$. $\square$

With Lemma 2 in place, we characterize the convergence rate of the ODE (25) in Theorem 2 below.

Theorem 2. *Let Assumption 1 hold, and X be the solution of (25) with $\alpha \leq 2$ and $p + \alpha \geq 2$. Then for any $T > T_0 > 0$,*

$$f(X(T)) - f^* \leq \frac{\mathcal{E}(T_0)}{2(p+1)T^{2-\alpha}}.$$

Proof. Lemma 2 implies that $\mathcal{E}(T) \leq \mathcal{E}(T_0)$ for $T \geq T_0$. Therefore, we have

$$f(X(T)) - f^* \leq \frac{\mathcal{E}(T)}{2(p+1)T^{2-\alpha}} \leq \frac{\mathcal{E}(T_0)}{2(p+1)T^{2-\alpha}},$$

and the proof is completed. $\square$

3 Convergence Results of SGDM

In this section, we carry our continuous-time constructions over to the analysis of the discrete-time case. In this connection, we consider the following SGDM method

$$x_{k+1} = x_k + \frac{k}{k+2}(x_k - x_{k-1}) - \frac{2\sqrt{\eta_k}}{(k+2)\sqrt{k}}g(x_k, \xi_k). \quad (28)$$

The scheme (28) is obtained from (1) by subscribing the first equality to the second equality and replacing η by η_k . The choice of stepsize η_k is presented below that does not rely on knowledge of the total number of iterates or the noise variance.

We use $\mathcal{F}_k$ to denote the σ -algebra generated by all randomness after arriving at x_k , that is, $\mathcal{F}_k = \sigma(\xi_1, \dots, \xi_{k-1})$. We introduce the following assumption for the discrete-time convergence analysis.

Assumption 2. *For any $k \geq 1$, the stochastic gradient $g(x_k, \xi_k)$ satisfies $\mathbb{E}[g(x_k, \xi_k)|\mathcal{F}_k] = \nabla f(x_k)$ and $\mathbb{E}[\|g(x_k, \xi_k)\|^2|\mathcal{F}_k] \leq \|\nabla f(x_k)\|^2 + \sigma^2$.*

3.1 Discrete-Time Lyapunov Function

Now we show how to translate the Lyapunov argument in Section 2 to the discrete-time case. Firstly, we define the discrete-time Lyapunov function

$$\mathcal{E}(k) = \|x_{k+1} + (k+1)(x_{k+1} - x_k) - x^*\|^2 + 4\sqrt{(k+1)\eta_k}(f(x_k) - f^*), \quad (29)$$

which is obtained from (26) by replacing t with $(k+1)\eta_k$, and $\dot{X}$ with $v_{k+1} = \frac{x_{k+1} - x_k}{\eta_k}$. The following result shows the descent property of the Lyapunov function $\mathcal{E}(k)$, and the proof of Lemma 3 is translated from that of Lemma 2 in the same way as the construction of $\mathcal{E}(k)$.

Lemma 3. *Let Assumption 1 hold. If the stepsize sequence $\{\eta_k\}$ is monotonically decreasing, then $\{x_k\}$ generated by SGDM (28) satisfies*

$$\begin{aligned} \mathcal{E}(k) - \mathcal{E}(k-1) &\leq \frac{4\eta_k}{k} \|g(x_k, \xi_k)\|^2 - \frac{2}{L} \sqrt{\frac{\eta_k}{k}} \|\nabla f(x_k)\|^2 - 2\sqrt{\frac{\eta_k}{k}}(f(x_k) - f^*) \\ &\quad + 4\sqrt{\frac{\eta_k}{k}} \langle \nabla f(x_k) - g(x_k, \xi_k), \tau_k \rangle \end{aligned} \quad (30)$$

for any $k \geq 1$, where $\tau_k = k(x_k - x_{k-1}) + (x_k - x^*)$.

We note that the first term $\frac{4\eta_k}{k} \|g(x_k, \xi_k)\|^2$ on the RHS of (30) is a higher-order term with respect to η_k than the second term $\frac{2}{L} \sqrt{\frac{\eta_k}{k}} \|\nabla f(x_k)\|^2$; so this positive term can be controlled when η_k is small. Next, the last term $4\sqrt{\frac{\eta_k}{k}} \langle \nabla f(x_k) - g(x_k, \xi_k), \tau_k \rangle$ has expectation equal to zero. Hence, $\mathcal{E}(k)$ trends downward with respect to k , making Lemma 3 the discrete-time counterpart of Lemma 2.

Proof of Lemma 3. By differencing the Lyapunov function (29), we get

$$\begin{aligned}
& \mathcal{E}(k) - \mathcal{E}(k-1) \\
& \leq \|x_{k+1} + (k+1)(x_{k+1} - x_k) - x^*\|^2 + 4\sqrt{(k+1)\eta_k}(f(x_k) - f^*) \\
& \quad - \|x_k + k(x_k - x_{k-1}) - x^*\|^2 - 4\sqrt{k\eta_k}(f(x_{k-1}) - f^*) \\
& \leq 2\langle 2(x_{k+1} - x_k) + k(x_{k+1} - 2x_k + x_{k-1}), x_{k+1} + (k+1)(x_{k+1} - x_k) - x^* \rangle \\
& \quad - \|2(x_{k+1} - x_k) + k(x_{k+1} - 2x_k + x_{k-1})\|^2 + 4\sqrt{k\eta_k}(f(x_k) - f(x_{k-1})) \\
& \quad + 2\sqrt{\frac{\eta_k}{k}}(f(x_k) - f^*),
\end{aligned} \tag{31}$$

where the first inequality follows from $\eta_k \leq \eta_{k-1}$, and the second inequality follows from $\|a\|^2 - \|b\|^2 = 2\langle a - b, a \rangle - \|a - b\|^2$ and $\sqrt{k+1} - \sqrt{k} \leq \frac{1}{2\sqrt{k}}$. From (28) we have

$$\begin{aligned}
& 2(x_{k+1} - x_k) + k(x_{k+1} - 2x_k + x_{k-1}) \\
& = (k+2)(x_{k+1} - x_k) - k(x_k - x_{k-1}) = -2\sqrt{\frac{\eta_k}{k}}g(x_k, \xi_k),
\end{aligned} \tag{32}$$

and thus

$$\begin{aligned}
& \mathcal{E}(k) - \mathcal{E}(k-1) \\
& \leq -4 \left\langle \sqrt{\frac{\eta_k}{k}}g(x_k, \xi_k), x_k + (k+2)(x_{k+1} - x_k) - x^* \right\rangle - \left\| 2\sqrt{\frac{\eta_k}{k}}g(x_k, \xi_k) \right\|^2 \\
& \quad + 4\sqrt{k\eta_k}(f(x_k) - f(x_{k-1})) + 2\sqrt{\frac{\eta_k}{k}}(f(x_k) - f^*) \\
& = -4 \left\langle \sqrt{\frac{\eta_k}{k}}g(x_k, \xi_k), (k+2)(x_{k+1} - x_k) \right\rangle - \left\| 2\sqrt{\frac{\eta_k}{k}}g(x_k, \xi_k) \right\|^2 \\
& \quad + 4\sqrt{k\eta_k}(f(x_k) - f(x_{k-1})) + 4\sqrt{\frac{\eta_k}{k}}(f(x_k) - f^* - \langle g(x_k, \xi_k), x_k - x^* \rangle) \\
& \quad - 2\sqrt{\frac{\eta_k}{k}}(f(x_k) - f^*).
\end{aligned} \tag{33}$$

Intuitively, (31) can be seen as the discrete-time counterpart of (27), and (32) corresponds to the ODE (25). Then we note that the convexity and L -smoothness of f gives

$$f^* - f(x_k) - \langle \nabla f(x_k), x^* - x_k \rangle \geq \frac{1}{2L} \|\nabla f(x_k)\|^2, \tag{34}$$

and the convexity of f shows that

$$f(x_k) - f(x_{k-1}) \leq \langle \nabla f(x_k), x_k - x_{k-1} \rangle. \tag{35}$$

Plugging (34) and (35) into (33), we have

$$\begin{aligned}
& \mathcal{E}(k) - \mathcal{E}(k-1) \\
& \leq 4\sqrt{\frac{\eta_k}{k}} \langle g(x_k, \xi_k), -(k+2)x_{k+1} + (2k+2)x_k - kx_{k-1} \rangle \\
& \quad + 4\sqrt{\frac{\eta_k}{k}} \langle \nabla f(x_k) - g(x_k, \xi_k), k(x_k - x_{k-1}) \rangle \\
& \quad - \left\| 2\sqrt{\frac{\eta_k}{k}} g(x_k, \xi_k) \right\|^2 - \frac{2}{L} \sqrt{\frac{\eta_k}{k}} \|\nabla f(x_k)\|^2 + 4\sqrt{\frac{\eta_k}{k}} \langle \nabla f(x_k) - g(x_k, \xi_k), x_k - x^* \rangle \\
& \quad - 2\sqrt{\frac{\eta_k}{k}} (f(x_k) - f^*).
\end{aligned}$$

Finally, we substitute (32) to the RHS of the above inequality, and reorder the terms to conclude

$$\begin{aligned}
\mathcal{E}(k) - \mathcal{E}(k-1) & \leq \frac{4\eta_k}{k} \|g(x_k, \xi_k)\|^2 - \frac{2}{L} \sqrt{\frac{\eta_k}{k}} \|\nabla f(x_k)\|^2 - 2\sqrt{\frac{\eta_k}{k}} (f(x_k) - f^*) \\
& \quad + 4\sqrt{\frac{\eta_k}{k}} \langle \nabla f(x_k) - g(x_k, \xi_k), \tau_k \rangle,
\end{aligned}$$

where $\tau_k = k(x_k - x_{k-1}) + p(x_k - x^*)$. □

3.2 Convergence in Expectation of SGDM

With Lemma 3 in place, we present the convergence result in expectation of SGDM. Furthermore, in the following analysis we show that there exists a subsequence of $\{x_k\}$ converges at rate $o\left(\frac{1}{\sqrt{k} \log \log k}\right)$, which is faster than the lower bound $\Omega\left(\frac{1}{\sqrt{k}}\right)$ of the whole sequence (Agarwal et al., 2012).

Theorem 3. *Let Assumptions 1 and 2 hold. Then $\{x_k\}$ generated by SGDM (28) with stepsize $\eta_k = \frac{1}{L^2 \log^2(k+2)}$ and initialization $x_0 = x_1$ satisfies*

$$\mathbb{E}[f(x_k)] - f^* \leq \frac{\left(3L\|x_0 - x^*\|^2 + \frac{4\sigma^2}{L}\right) \log(k+2)}{2\sqrt{k+1}},$$

and

$$\liminf_{k \rightarrow \infty} \left(\sqrt{k} \log \log(k+2) \right) \left(\mathbb{E}[f(x_k)] - f^* \right) = 0. \tag{36}$$

Proof. Since τ_k is $\mathcal{F}_k$ -measurable, Assumption 2 gives

$$\mathbb{E}[\langle \nabla f(x_k) - g(x_k, \xi_k), \tau_k \rangle | \mathcal{F}_k] = 0$$

and

$$\mathbb{E}[\|g(x_k, \xi_k)\|^2 | \mathcal{F}_k] \leq \|\nabla f(x_k)\|^2 + \sigma^2.$$

By letting $\eta_k = \frac{1}{4L^2 \log^2(k+2)}$, we have $\frac{4\eta_k}{k} \leq \frac{2}{L} \sqrt{\frac{\eta_k}{k}}$ for any $k \geq 1$. Then we take conditional expectation with respect to $\mathcal{F}_k$ on both sides of (30) to get

$$\mathbb{E}[\mathcal{E}(k) - \mathcal{E}(k-1) | \mathcal{F}_k] \leq \frac{4\eta_k}{k} \sigma^2 - 2\sqrt{\frac{\eta_k}{k}} (f(x_k) - f^*).$$

Consequently, taking expectation on both sides yields that

$$\mathbb{E}[\mathcal{E}(k)] - \mathbb{E}[\mathcal{E}(k-1)] \leq \frac{4\eta_k}{k}\sigma^2 - 2\sqrt{\frac{\eta_k}{k}}(\mathbb{E}[f(x_k)] - f^*), \quad (37)$$

for any $k \geq 1$. Summing (37) over k yields that

$$\mathbb{E}[\mathcal{E}(k)] \leq \mathcal{E}(0) + 4\sigma^2 \sum_{i=1}^k \frac{\eta_i}{i}. \quad (38)$$

Additionally, the definition of the Lyapunov function (29) gives $4\sqrt{(k+1)\eta_k}(\mathbb{E}[f(x_k)] - f^*) \leq \mathbb{E}[\mathcal{E}(k)]$ and

$$\mathcal{E}(0) = \|x_0 - x^*\|^2 + \frac{2}{\log 2 \cdot L}(f(x_0) - f^*) \leq \left(1 + \frac{2}{\log 2 \cdot L} \cdot \frac{L}{2}\right) \|x_0 - x^*\|^2 \leq 3\|x_0 - x^*\|^2.$$

Substituting these two inequalities to (38) implies that

$$\mathbb{E}[f(x_k)] - f^* \leq \frac{3\|x_0 - x^*\|^2 + \frac{\sigma^2}{L^2} \sum_{i=1}^k \frac{1}{i \log^2(i+2)}}{4\sqrt{\frac{k+1}{4L^2 \log^2(k+2)}}} \leq \frac{\left(3L\|x_0 - x^*\|^2 + \frac{4\sigma^2}{L}\right) \log(k+2)}{2\sqrt{k+1}},$$

where the second inequality follows from $\sum_{i=1}^{\infty} \frac{1}{i \log^2(i+2)} \leq 2 + \frac{1}{\log 2} \leq 4$. For the second part of the theorem, we rearrange (37) as

$$2\sqrt{\frac{\eta_k}{k}}(\mathbb{E}[f(x_k)] - f^*) \leq \mathbb{E}[\mathcal{E}(k-1)] - \mathbb{E}[\mathcal{E}(k)] + \frac{4\eta_k}{k}\sigma^2,$$

and then sum over k to get

$$\begin{aligned} \sum_{i=1}^k \frac{1}{L} \sqrt{\frac{1}{i \log^2(i+2)}} (\mathbb{E}[f(x_i)] - f^*) &\leq \mathcal{E}(0) - \mathbb{E}[\mathcal{E}(k)] + \frac{\sigma^2}{L^2} \sum_{i=1}^k \frac{1}{i \log^2(i+2)} \\ &\leq \mathcal{E}(0) + \frac{\sigma^2}{L^2} \sum_{i=1}^{\infty} \frac{1}{i \log^2(i+2)}. \end{aligned} \quad (39)$$

The RHS of (39) is a finite number, and we denote it by D for convenience. Suppose that there exists $\mu \geq 0$ and $N \geq 1$ such that $\mathbb{E}[f(x_i)] - f^* \geq \frac{\mu}{\sqrt{i \log \log(i+2)}}$ for any $i \geq N$, then we choose $k \geq N$ in (39), and the LHS is lower bounded by $\frac{\mu}{L} \sum_{i=N}^k \frac{1}{i \log(i+2) \log \log(i+2)}$. Substitute this inequality to (39) and take limit on both sides yield that

$$\frac{\mu}{L} \lim_{k \rightarrow \infty} \sum_{i=N}^k \frac{1}{i \log(i+2) \log \log(i+2)} \leq D,$$

which leads to a contradiction since

$$\lim_{k \rightarrow \infty} \sum_{i=N}^k \frac{1}{i \log(i+2) \log \log(i+2)} = \infty.$$

Therefore, we finish the proof of (36). $\square$

3.3 Almost Sure Convergence of SGDM

By using a classical supermartingale convergence theorem, Lemma 3 also leads to the following almost sure convergence result.

Theorem 4. *Let Assumptions 1 and 2 hold. If $\{r_k\}$ satisfies $\lim_{k \rightarrow \infty} r_k = 0$, then for $\{x_k\}$ generated by SGDM (28) with stepsize $\eta_k = \frac{1}{L^2 \log^2(k+2)}$ and initialization $x_0 = x_1$, we have almost surely that*

$$\lim_{k \rightarrow \infty} r_k \cdot \frac{\sqrt{k}}{\log k} \cdot (f(x_k) - f^*) = 0,$$

and

$$\liminf_{k \rightarrow \infty} \left(\sqrt{k} \log \log(k+2) \right) (f(x_k) - f^*) = 0.$$

Proof. The proof of Theorem 4 relies on the following supermartingale convergence result in Robbins and Siegmund (1971).

Lemma 4. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \dots$ a sequence of sub- σ -algebras of $\mathcal{F}$. For each $k = 1, 2, \dots$, let z_k , ξ_k and ζ_k be non-negative $\mathcal{F}_k$ -measurable random variables such that $\sum_{i=1}^{\infty} \xi_i < \infty$ almost surely and*

$$\mathbb{E}[z_{k+1} | \mathcal{F}_k] \leq z_k + \xi_k - \zeta_k,$$

for any $k \geq 1$. Then $\lim_{k \rightarrow \infty} z_k$ exists and is finite and $\sum_{i=0}^{\infty} \zeta_i < \infty$ almost surely.

In the proof of Theorem 3, we have shown that with stepsize $\eta_k = \frac{1}{L^2 \log^2(k+2)}$, the inequality

$$\mathbb{E}[\mathcal{E}(k) - \mathcal{E}(k-1) | \mathcal{F}_k] \leq \frac{4\eta_k}{k} \sigma^2 - 2\sqrt{\frac{\eta_k}{k}} (f(x_k) - f^*)$$

holds for any $k \geq 1$. The definition yields that $\mathcal{E}(k-1)$ and $2\sqrt{\frac{\eta_k}{k}} (f(x_k) - f^*)$ are $\mathcal{F}_k$ -measurable, and $\sum_{i=1}^{\infty} \frac{4\eta_i}{i} \sigma^2 < \infty$. Therefore, we let $z_k = \mathcal{E}(k-1)$, $\xi_k = \frac{4\eta_k}{k} \sigma^2$, and $\zeta_k = 2\sqrt{\frac{\eta_k}{k}} (f(x_k) - f^*)$, and thus applying Lemma 4 gives

1. $\lim_{k \rightarrow \infty} \mathcal{E}(k) = \mathcal{E}_{\infty}$ exists with probability one;
2. $\sum_{k=1}^{\infty} \sqrt{\frac{\eta_k}{k}} (f(x_k) - f^*)$ exists with probability one.

The definition of the Lyapunov function (29) yields that

$$r_k \cdot \frac{\sqrt{k}}{\log k} \cdot (f(x_k) - f^*) \leq r_k \cdot \frac{\sqrt{k}}{\log k} \cdot \frac{\mathcal{E}(k)}{4\sqrt{(k+1)\eta_k}} = r_k \cdot \frac{\sqrt{k}}{\log k} \cdot \frac{L \log(k+2)}{4\sqrt{(k+1)}} \cdot \mathcal{E}(k). \quad (40)$$

Since $\lim_{k \rightarrow \infty} \mathcal{E}(k)$ exists almost surely and $\lim_{k \rightarrow \infty} r_k = 0$, (40) yields that $\lim_{k \rightarrow \infty} r_k \cdot \frac{\sqrt{k}}{\log k} \cdot (f(x_k) - f^*) = 0$ almost surely, and thus we finish the proof of the first part.

For the second part of the theorem, we would show that

$$\sum_{k=1}^{\infty} \sqrt{\frac{\eta_k}{k}} (f(x_k) - f^*) < \infty \quad (41)$$

yields

$$\liminf_{k \rightarrow \infty} \frac{f(x_k) - f^*}{\frac{1}{\sqrt{k} \log \log(k+2)}} = 0. \quad (42)$$

Suppose that there exist $\mu \geq 0$ and $N \geq 1$ such that $f(x_i) - f^* \geq \frac{\mu}{\sqrt{i} \log \log(i+2)}$ for any $i \geq N$. Then we have

$$\sum_{k=1}^{\infty} \sqrt{\frac{\eta_k}{k}} (f(x_k) - f^*) \geq \frac{\mu}{L} \sum_{k=N}^{\infty} \frac{1}{k \log(k+2) \log \log(k+2)} = \infty,$$

which contradicts (41). Since (41) holds almost surely, this contradiction yields that (42) also holds almost surely, and we complete the proof of Theorem 4. $\square$

4 Anytime High Probability Convergence Result

In this section we present the anytime high probability convergence result for SGDM. Our analysis is based on the dynamical property illustrated by the discrete Lyapunov function $\mathcal{E}(k)$ and Lemma 3. We introduce a new approach to bound the increment $\mathcal{E}(k) - \mathcal{E}(k-1)$ by $\mathcal{E}(k-1)$, the Lyapunov function of the previous step, which leads to a Gronwall-type argument. This technique is critical in removing the reliance on additional projection steps or assumptions of bounded gradients or stochastic gradients. Subsequently, by modifying the moment generating function of $\mathcal{E}(k)$, we construct a supermartingale and establish the desired anytime high probability convergence. To begin with, we introduce the following assumption.

Assumption 3. For any $k \geq 1$, the stochastic gradient $g(x_k, \xi_k)$ satisfies

$$\mathbb{E} \left[\exp \left(\frac{\|g(x_k, \xi_k) - \nabla f(x_k)\|^2}{\sigma^2} \right) \middle| \mathcal{F}_k \right] \leq \exp(1).$$

We now present our main convergence result.

Theorem 5. Let Assumptions 1, 2 and 3 hold. For any β , the sequence $\{x_k\}$ generated by SGDM (28) with stepsize $\eta_k = \frac{1}{16L^2 \log^2(k+2)}$ and initialization $x_0 = x_1$ satisfies

$$\mathbb{P} \left(f(x_k) - f^* \leq \frac{(C_1 + C_2 \log \frac{1}{\beta}) \log(k+2)}{\sqrt{k+1}}, \text{ for all } k \geq 0 \right) \geq 1 - 2\beta,$$

where $C_1 = L\gamma_2\mathcal{E}(0) + L\sigma^2(1 + \sigma^2\gamma_1\gamma_2)\gamma_1$, $C_2 = L\gamma_2 + L\sigma^2(1 + \sigma^2\gamma_1\gamma_2)\gamma_1$, $\gamma_1 = \sum_{k=1}^{\infty} \frac{16\eta_k}{k}$, and $\gamma_2 = \prod_{k=1}^{\infty} (1 + \frac{16\eta_k}{k}\sigma^2)$.

Before proceeding with the proof, we explain the intuition of our approach. The Lyapunov function $\mathcal{E}(k)$ and Lemma 3 provides the dynamical property of SGDM. More specifically, Lemma 3 yields that

$$\begin{aligned} \mathcal{E}(k) - \mathcal{E}(k-1) &\leq \frac{4\eta_k}{k} \|g(x_k, \xi_k)\|^2 - \frac{2}{L} \sqrt{\frac{\eta_k}{k}} \|\nabla f(x_k)\|^2 + 4\sqrt{\frac{\eta_k}{k}} \langle \nabla f(x_k) - g(x_k, \xi_k), \tau_k \rangle \\ &\leq \frac{8\eta_k}{k} \|\theta_k\|^2 + \frac{8\eta_k}{k} \|\nabla f(x_k)\|^2 - \frac{2}{L} \sqrt{\frac{\eta_k}{k}} \|\nabla f(x_k)\|^2 + 4\sqrt{\frac{\eta_k}{k}} \langle \theta_k, \tau_k \rangle \\ &\leq a_k \|\theta_k\|^2 + \sqrt{a_k} \langle \theta_k, \tau_k \rangle, \end{aligned} \quad (43)$$

where $\tau_k = k(x_k - x_{k-1}) + (x_k - x^*)$, $\theta_k = \nabla f(x_k) - g(x_k, \xi_k)$, $a_k = \frac{16\eta_k}{k}$, and the second inequality follows from $\eta_k \leq \frac{k}{16L^2}$ for $k \geq 1$. Intuitively, the above inequality (43) decouples the increment of $\mathcal{E}(k)$ into two terms. Classic concentration arguments bound the sum of the first part, $\sum_k a_k \|\theta_k\|^2$ with high probability. The second part, however, is more involved. Noticing $\|\tau_k\|^2 \leq \mathcal{E}(k-1)$, we derive a Gronwall-type inequality and construct a supermartingale by modifying the moment generating function of $\mathcal{E}(k)$. Consequently, we derive the desired anytime convergence result. To simplify the proof, we assume the stochastic terms $\{\xi_k\}$ are independent in the analysis below. This simplification can be removed by the martingale argument using conditional expectation.

Proof. To prove Theorem 5, we would present concentration estimates on $\mathbb{P}(\sup_{k \geq 1} \mathcal{E}(k) \geq \alpha)$. We denote $S(k) = \sum_{l=1}^k a_l \|\theta_l\|^2$ and $M(k) = \mathcal{E}(k) - S(k)$. This gives

$$M(k) - M(k-1) = (\mathcal{E}(k) - \mathcal{E}(k-1)) - (S(k) - S(k-1)) \leq \sqrt{a_k} \langle \theta_k, \tau_k \rangle. \quad (44)$$

We write $\gamma_1 = \sum_{k=1}^{\infty} a_k$, and $\gamma_2 = \prod_{k=1}^{\infty} (1 + a_k \sigma^2)$. It is direct to verify that these two constants are finite for $a_k = \frac{16\eta_k}{k} = \frac{1}{L^2 k \log^2(k+2)}$.

For the moment-generating function of $M(k)$, (44) gives

$$\mathbb{E}[\exp(tM(k)) | \mathcal{F}_{k-1}] \leq \exp(tM(k-1)) \mathbb{E}[\exp(t\sqrt{a_k} \langle \theta_k, \tau_k \rangle) | \mathcal{F}_{k-1}].$$

Definition of the Lyapunov function (29) yields that $\|\tau_k\|^2 \leq \mathcal{E}(k-1)$, and thus Lemma 5 gives

$$\mathbb{E}[\exp(t\sqrt{a_k}\langle\theta_k, \tau_k\rangle)|\mathcal{F}_{k-1}] \leq \exp(a_k\sigma^2 t^2 \mathcal{E}(k-1)).$$

Therefore, for $t \leq 1$, we have

$$\begin{aligned} \mathbb{E}[\exp(tM(k))|\mathcal{F}_{k-1}] &\leq \exp(tM(k-1) + a_k\sigma^2 t^2 \mathcal{E}(k-1)) \\ &\leq \exp((1 + a_k\sigma^2)tM(k-1) + a_k\sigma^2 tS(k-1)), \end{aligned} \quad (45)$$

where the second inequality follows from $\mathcal{E}(k-1) = M(k-1) + S(k-1)$ and $t \leq 1$.

We write

$$N^t(k) = \exp\left(\prod_{l=k+1}^{\infty} (1 + a_l\sigma^2)tM(k) - \sum_{l=1}^k a_l\sigma^2\gamma_2 tS(l-1)\right).$$

Specifically, this definition gives $N^t(0) = \exp(\prod_{l=0}^{\infty}(1 + a_l\sigma^2)tM(0)) = \exp(\gamma_2 t\mathcal{E}(0))$. The conditional expectation of $N^t(k)$ for $t \leq \frac{1}{\gamma_2}$ satisfies

$$\begin{aligned} &\mathbb{E}[N^t(k)|\mathcal{F}_{k-1}] \\ &= \mathbb{E}\left[\exp\left(\prod_{l=k+1}^{\infty} (1 + a_l\sigma^2)tM(k) - \sum_{l=1}^k a_l\sigma^2\gamma_2 tS(l-1)\right)\middle|\mathcal{F}_{k-1}\right] \\ &= \mathbb{E}\left[\exp\left(\prod_{l=k+1}^{\infty} (1 + a_l\sigma^2)tM(k)\right)\middle|\mathcal{F}_{k-1}\right] \cdot \exp\left(-\sum_{l=1}^k a_l\sigma^2\gamma_2 tS(l-1)\right) \\ &\leq \exp\left(\prod_{l=k}^{\infty}(1 + a_l\sigma^2)tM(k-1) + a_k\sigma^2 \prod_{l=k+1}^{\infty} (1 + a_l\sigma^2)tS(k-1) - \sum_{l=1}^k a_l\sigma^2\gamma_2 tS(l-1)\right) \\ &= N^t(k-1) \cdot \exp\left(a_k\sigma^2 \prod_{l=k+1}^{\infty} (1 + a_l\sigma^2)tS(k-1) - a_k\sigma^2\gamma_2 tS(k-1)\right) \\ &\leq N^t(k-1), \end{aligned} \quad (46)$$

where (46) follows from (45). Therefore, $\{N^t(k)\}_{k=0}^{\infty}$ is a supermartingale for $0 < t \leq \frac{1}{\gamma_2}$. Then applying Ville's inequality (Ville, 1939) to $\{N^t(k)\}_{k=0}^{\infty}$ gives

$$\mathbb{P}\left(\sup_{k \geq 0} N^t(k) \geq \exp(\alpha t)\right) \leq \exp(-\alpha t)\mathbb{E}[N^t(0)] = \exp(-\alpha t + \gamma_2 t\mathcal{E}(0)). \quad (47)$$

Since $\gamma_1 S(k) \geq \sum_{l=1}^k a_l S(l-1)$, we have

$$\exp\left(\frac{M(k)}{\gamma_2} - \sigma^2\gamma_1 S(k)\right) \leq \exp\left(\frac{\prod_{l=k+1}^{\infty}(1 + a_l\sigma^2)M(k)}{\gamma_2} - \sum_{l=1}^k a_l\sigma^2 S(l-1)\right) = N^{\frac{1}{\gamma_2}}(k),$$

and thus

$$\begin{aligned} &\mathbb{P}\left(\sup_{k \geq 0} \{M(k) - \sigma^2\gamma_1\gamma_2 S(k)\} \geq \gamma_2 \left(\mathcal{E}(0) + \log \frac{1}{\beta}\right)\right) \\ &= \mathbb{P}\left(\sup_{k \geq 0} \left\{\exp\left(\frac{M(k)}{\gamma_2} - \sigma^2\gamma_1 S(k)\right)\right\} \geq \exp\left(\mathcal{E}(0) + \log \frac{1}{\beta}\right)\right) \\ &\leq \mathbb{P}\left(\sup_{k \geq 0} N^{\frac{1}{\gamma_2}}(k) \geq \exp\left(\mathcal{E}(0) + \log \frac{1}{\beta}\right)\right) \leq \beta, \end{aligned} \quad (48)$$

where the last inequality follows from (47) with $t = \frac{1}{\gamma_2}$ and $\alpha = \gamma_2 \left(\mathcal{E}(0) + \log \frac{1}{\beta} \right)$.

On the other hand, Lemma 6 gives

$$\begin{aligned} & \mathbb{P} \left((1 + \sigma^2 \gamma_1 \gamma_2) S(k) \geq \left(1 + \log \frac{1}{\beta} \right) \sigma^2 (1 + \sigma^2 \gamma_1 \gamma_2) \gamma_1 \right) \\ & \leq \mathbb{P} \left((1 + \sigma^2 \gamma_1 \gamma_2) S(k) \geq \left(1 + \log \frac{1}{\beta} \right) \sigma^2 (1 + \sigma^2 \gamma_1 \gamma_2) \sum_{l=1}^k a_l \right) \leq \beta. \end{aligned}$$

Since $\{S(k)\}_{k=0}^\infty$ is monotonically increasing, we further have

$$\mathbb{P} \left(\sup_{k \geq 0} \{ (1 + \sigma^2 \gamma_1 \gamma_2) S(k) \} \geq \left(1 + \log \frac{1}{\beta} \right) \sigma^2 (1 + \sigma^2 \gamma_1 \gamma_2) \gamma_1 \right) \leq \beta. \quad (49)$$

Combining (48) and (49), we derive

$$\begin{aligned} & \mathbb{P} \left(\sup_{k \geq 0} \mathcal{E}(k) \geq \gamma_2 \left(\mathcal{E}(0) + \log \frac{1}{\beta} \right) + \left(1 + \log \frac{1}{\beta} \right) \sigma^2 (1 + \sigma^2 \gamma_1 \gamma_2) \gamma_1 \right) \\ & \leq \mathbb{P} \left(\sup_{k \geq 0} \{ M(k) - \sigma^2 \gamma_1 \gamma_2 S(k) \} \geq \gamma_2 \left(\mathcal{E}(0) + \log \frac{1}{\beta} \right) \right) \\ & \quad + \mathbb{P} \left(\sup_{k \geq 0} \{ (1 + \sigma^2 \gamma_1 \gamma_2) S(k) \} \geq \left(1 + \log \frac{1}{\beta} \right) \sigma^2 (1 + \sigma^2 \gamma_1 \gamma_2) \gamma_1 \right) \\ & \leq 2\beta. \end{aligned}$$

Since $4\sqrt{(k+1)\eta_k}(f(x_k) - f^*) \leq \mathcal{E}(k)$, we conclude that

$$\mathbb{P} \left(f(x_k) - f^* \leq \frac{\left(C_1 + C_2 \log \frac{1}{\beta} \right) \log(k+2)}{\sqrt{k+1}}, \text{ for all } k \geq 0 \right) \geq 1 - 2\beta,$$

where $C_1 = L\gamma_2\mathcal{E}(0) + L\sigma^2(1 + \sigma^2\gamma_1\gamma_2)\gamma_1$, and $C_2 = L\gamma_2 + L\sigma^2(1 + \sigma^2\gamma_1\gamma_2)\gamma_1$. $\square$

Remark 1. By slightly modifying the stepsize, the convergence rate in Theorem 5 can be improved to $\mathcal{O} \left(\frac{\log \frac{1}{\beta} \log \frac{1+\varepsilon}{2} k}{\sqrt{k}} \right)$, where ε is a positive number that can be arbitrarily small; see Appendix D for more details.

5 Experiments

In this section, we test SGDM, in comparison with SGD, on an artificial example, a logistic regression problem, and two image classification tasks.

Firstly, we investigate the performance of SGDM on an artificial optimization problem and a logistic regression task. In the artificial experiment, the objective is a 10-dimensional quadratic function, and a Gaussian noise is added to each gradient output. In the logistic regression task, the dataset is taken from LIBSVM library (Chang and Lin, 2011), and the approximation of the minimal function value f^* is obtained by a standard solver from `scipy` (Virtanen et al., 2020). For SGD, we use the classic stepsize $\eta_k = \frac{1}{\sqrt{k}}$. The results are presented in Figure 2(a) and 2(b). We observe that SGDM converges faster than SGD. Furthermore, the gradient noise in each iteration of SGDM is multiplied by a factor of order $\tilde{\mathcal{O}} \left(\frac{1}{k^{\frac{3}{2}}} \right)$, while that of SGD is multiplied by a factor of order $\mathcal{O} \left(\frac{1}{\sqrt{k}} \right)$. This results in the error curve of SGDM being significantly smoother than SGD, as observed from the individual run in the subplot of Figure 2(a) and 2(b).

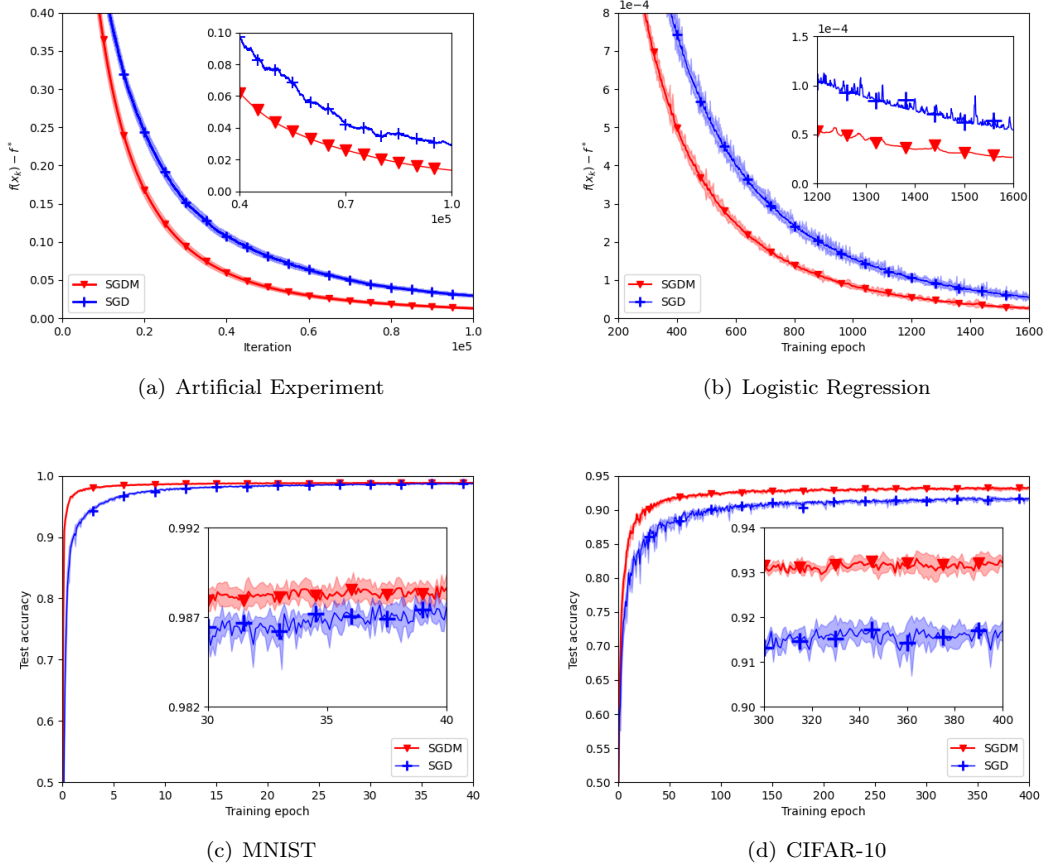


Figure 2: Results of the artificial experiment (Figure 2(a)), logistic regression (Figure 2(b)), image classification on MNIST (Figure 2(c)) and CIFAR-10 (Figure 2(d)). The solid lines in the main plot of all figures are obtained through averaging 10 independent runs, and the variance regions are represented by shaded areas above and below the solid lines. The subplot in Figure 2(a) and 2(b) presents an individual run, and the subplots in Figure 2(c) and 2(d) are zoomed in to provide better views of the last 1/4 training processes.

Detailed settings about these two experiments and more plots of individual runs are presented in Appendix E.

Then we apply both SGDM and SGD to train neural networks for two image classification tasks, MNIST and CIFAR-10. The model used in the MNIST task is a two-layer CNN, and the model used in the CIFAR-10 task is Resnet-18 (He et al., 2016). The learning rate of SGD is tuned by the common hyperparameter selection practice (Bergstra and Bengio, 2012). More details can be found in Appendix E. The results are presented in Figure 2(c) and 2(d), which confirm the better performance of SGDM. Comparison of individual runs (presented in Appendix E) also shows that the learning curves of SGDM have smaller oscillations than SGD.

6 Conclusion

In this paper, we study the classic problem of stochastic first-order method for smooth convex programs. Despite numerous existing works, several fundamental questions remain unanswered. Among many open problems, one of the most important is:

Can we derive an anytime convergence rate without losing any factors in k , compared to the state-of-the-art convergence rate for a single-iterate guarantee?

This paper gives an affirmative answer to this above question. This result is achieved by a careful analysis of the dynamics of stochastic gradient with momentum (SGDM). In particular, we establish a continuous-discrete time correspondence for SGDMs by a rigorous stochastic analysis.

This work paves the way for exploring possibilities in anytime analysis across various settings. For instance, it would be intriguing to investigate the potential establishment of a similar anytime convergence rate for non-convex programs.

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A The continuous-time limit of the AC-SA algorithm (Lan, 2020)

In this section, we give an informal derivation of the continuous-time limit of the well-known AC-SA algorithm, and show that it is also of the form (2).

For smooth convex objective function f and Euclidean distance, the AC-SA algorithm (Lan, 2020) takes the form

$$\begin{aligned} y_k &= (1 - \alpha_k)x_{k-1} + \alpha_k z_{k-1}, \\ z_k &= z_{k-1} - \gamma_k g(y_k, \xi_k), \\ x_k &= (1 - \alpha_k)x_{k-1} + \alpha_k z_k, \end{aligned}$$

where the parameters are given as $\alpha_k = \frac{2}{k+1}$ and $\frac{1}{\gamma_k} = \frac{2L}{k} + \gamma\sqrt{k}$ (Lan, 2020, Proposition 4.5). For sufficiently large k , the latter is the dominant term, so we approximately set $\gamma_k = \frac{1}{\gamma\sqrt{k}}$. Furthermore, we take $\eta = \frac{1}{\gamma^2}$ and introduce the continuous-discrete connection between $X(k\eta)$ and x_k , which plays the same role as the *ansatz* $x_k \approx X(k\sqrt{s})$ in Su et al. (2014). As in the Theorem 1, we let $g(y_k, \xi_k) = \nabla f(y_k) + \xi_k$ for each k , where $\xi_k \stackrel{iid}{\sim} \mathcal{N}(0, I_n)$. Substituting the above parameters, the AC-SA scheme becomes

$$y_k = \frac{k-1}{k+1}x_{k-1} + \frac{2}{k+1}z_{k-1}, \tag{50}$$

$$z_k = z_{k-1} - \frac{\sqrt{\eta}}{\sqrt{k}}(\nabla f(y_k) + \xi_k), \tag{51}$$

$$x_k = \frac{k-1}{k+1}x_{k-1} + \frac{2}{k+1}z_k. \tag{52}$$

Rewriting (52) as

$$z_k = \frac{k+1}{2}x_k - \frac{k-1}{2}x_{k-1} \quad (53)$$

and substituting (50) and (53) to (51) yields that

$$\begin{aligned} & \frac{k+1}{2}x_k - \frac{k-1}{2}x_{k-1} \\ &= \frac{k}{2}x_{k-1} - \frac{k-2}{2}x_{k-2} - \frac{\sqrt{\eta}}{\sqrt{k}} \left(\nabla f \left(\frac{k}{k+1}x_k + \frac{k-1}{k+1}x_{k-1} - \frac{k-2}{k+1}x_{k-2} \right) + \xi_k \right), \end{aligned}$$

and thus

$$x_k - x_{k-1} = \frac{k-2}{k+1}(x_{k-1} - x_{k-2}) - \frac{2\sqrt{\eta}}{(k+1)\sqrt{k}} \left(\nabla f \left(x_k - \frac{1}{k+1}(x_k - x_{k-1}) + \frac{k-2}{k+1}(x_{k-1} - x_{k-2}) \right) + \xi_k \right).$$

We let $v_k = \frac{x_k - x_{k-1}}{\eta}$, then the above equality gives

$$v_k = \frac{k-2}{k+1}v_{k-1} - \frac{2}{(k+1)\sqrt{k\eta}} \left(\nabla f \left(x_k - \frac{\eta}{k+1}v_k + \frac{(k-2)\eta}{k+1}v_{k-1} \right) + \xi_k \right). \quad (54)$$

Taylor expansion of ∇f yields that

$$\nabla f \left(x_k - \frac{\eta}{k+1}v_k + \frac{(k-2)\eta}{k+1}v_{k-1} \right) = \nabla f(x_k) + \nabla^2 f(\theta_k) \left(-\frac{1}{k+1}v_k + \frac{k-2}{k+1}v_{k-1} \right) \eta,$$

where $\theta_k = \theta_k \left(x_k, x_k - \frac{\eta}{k+1}v_k + \frac{(k-2)\eta}{k+1}v_{k-1} \right)$. By reorganizing (54) and substituting the above Taylor expansion, the AC-SA scheme becomes

$$\begin{cases} x_k - x_{k-1} = \eta v_k, \\ v_k - v_{k-1} = -\frac{3}{(k-2)\eta}v_k\eta - \frac{2}{(k-2)\eta\sqrt{k\eta}}\nabla f(x_k)\eta - \frac{2}{(k-2)\eta\sqrt{k\eta}}(\sqrt{\eta}\xi_k)\sqrt{\eta} \\ \quad - \frac{2}{(k-2)\eta\sqrt{k\eta}}\nabla^2 f(\theta_k) \left(\frac{(k-3)\eta}{(k+1)\eta}v_k - \frac{(k-2)\eta}{(k+1)\eta}(v_k - v_{k-1}) \right) \eta^2. \end{cases} \quad (55)$$

Let W be a standard Brownian motion. As in Section 2.1, we let η in (55) go to zero and replace x_k , v_k , θ_k , $x_k - x_{k-1}$, $v_k - v_{k-1}$, $k\eta$, η and $\sqrt{\eta}\xi$ by X , V , Θ , dX , dV , t , dt and dW , respectively. The discrete scheme (55) becomes

$$\begin{cases} dX = Vdt, \\ dV = -\frac{3}{t}Vdt - \frac{2}{t^{\frac{3}{2}}}\nabla f(X)dt - \frac{2}{t^{\frac{3}{2}}}\sqrt{dt}dW - \frac{2}{t^{\frac{3}{2}}}\nabla^2 f(\Theta)(V - dV)(dt)^2. \end{cases}$$

Then we omit the high order terms and obtain

$$\begin{cases} dX = Vdt, \\ dV = -\frac{3}{t}Vdt - \frac{2}{t^{\frac{3}{2}}}\nabla f(X)dt. \end{cases}$$

Consequently, substituting the first line to the second line yields that

$$\ddot{X}(t) + \frac{3}{t}\dot{X}(t) + \frac{2}{t^{3/2}}\nabla f(X(t)) = 0,$$

which is also of the form (2).

B Additional related works

Stochastic optimization has a long history, with its origins dating back to at least (Robbins and Monro, 1951; Kiefer and Wolfowitz, 1952). In recent years, the field has experienced significant growth, driven by the widespread use of mini-batch training in machine learning (Goodfellow et al., 2016), which induces stochasticity. Researchers have approached the problem of stochastic optimization from various perspectives. For instance, Johnson and Zhang (2013), Allen-Zhu and Hazan (2016), Allen-Zhu (2017), Allen-Zhu (2018), and Ge et al. (2019) investigated algorithms that only use full-batch training sporadically for finite-sum optimization problems. This class of algorithms are commonly known as Stochastic Variance Reduced Gradient (SVRG). Li and Orabona (2019), Ward et al. (2020), and Faw et al. (2022) provided theoretical results for stochastic gradient methods with adaptive stepsizes. Zhang et al. (2020) demonstrated, both theoretically and empirically, that the heavy-tailed nature of gradient noise contributes to the advantage of adaptive gradient methods over SGD. Gadat et al. (2018); Kidambi et al. (2018); Gitman et al. (2019) studied the role of momentum in stochastic gradient methods. Karimi et al. (2016) and Madden et al. (2020) proved the convergence of stochastic gradient methods under the PL condition. Sebbouh et al. (2021) showed the almost sure convergence of SGD and the stochastic heavy-ball method. The convergence of stochastic gradient methods in expectation is also well studied. Shamir and Zhang (2013) studied the last-iterate convergence of SGD in expectation for non-smooth objective functions, and provided an optimal averaging scheme. Yan et al. (2018) presented a unified convergence analysis for stochastic momentum methods. Assran and Rabbat (2020) studied the convergence of Nesterov’s accelerated gradient method in stochastic settings. Liu et al. (2020) also provided an improved convergence analysis in expectation of a momentumized SGD, and proved the benefit of using the multistage strategy.

C Auxiliary lemmas

The proof of Theorem 5 relies on the following two auxiliary lemmas. These results have appeared in Lan et al. (2012); Devolder et al. (2011), and we include the proofs for completeness.

Lemma 5. *Let $\theta_1, \dots, \theta_k$ be a sequence of i.i.d. random variables, $\Gamma_l = \Gamma(\theta_{[l]})$ and $\Delta_l = \Delta(\theta_{[l]})$ be deterministic functions of $\theta_{[l]} = (\theta_1, \dots, \theta_l)$, and $c_1, \dots, c_k$ be a sequence of positive numbers such that:*

1. $\mathbb{E}[\Gamma_l | \theta_{[l-1]}] = 0$,
2. $|\Gamma_l| \leq c_l \Delta_l$,
3. $\mathbb{E} \left[\exp \left(\frac{\Delta_l^2}{\sigma^2} \right) \middle| \theta_{[l-1]} \right] \leq \exp(1)$,

hold for each $l \leq k$. Then, for any $\lambda \in \mathbb{R}$ and $l \geq 1$, we have

$$\mathbb{E} \left[\exp \left(\frac{\lambda \Gamma_l}{c_l \sigma} \right) \middle| \theta_{[l-1]} \right] \leq \exp \left(\frac{3\lambda^2}{4} \right).$$

Proof. Since $\exp(x) \leq x + \exp \left(\frac{9x^2}{16} \right)$ for any x , we have

$$\begin{aligned} \mathbb{E} \left[\exp \left(\frac{\lambda \Gamma_l}{c_l \sigma} \right) \middle| \theta_{[l-1]} \right] &\leq \mathbb{E} \left[\frac{\lambda \Gamma_l}{c_l \sigma} \middle| \theta_{[l-1]} \right] + \mathbb{E} \left[\exp \left(\frac{9\lambda^2 \Gamma_l^2}{16c_l^2 \sigma^2} \right) \middle| \theta_{[l-1]} \right] \\ &\leq \mathbb{E} \left[\exp \left(\frac{9\lambda^2 \Delta_l^2}{16\sigma^2} \right) \middle| \theta_{[l-1]} \right], \end{aligned}$$

for any λ , where the second inequality uses conditions 1 and 2. From the concavity of $f(x) = x^p$ for $0 < p \leq 1$, we further have

$$\mathbb{E} \left[\exp \left(\frac{\lambda \Gamma_l}{c_l \sigma} \right) \middle| \theta_{[l-1]} \right] \leq \left(\mathbb{E} \left[\exp \left(\frac{\Delta_l^2}{\sigma^2} \right) \middle| \theta_{[l-1]} \right] \right)^{\frac{9\lambda^2}{16}} \leq \exp \left(\frac{9\lambda^2}{16} \right), \quad (56)$$

for any $0 < \lambda \leq \frac{4}{3}$, where the second inequality uses condition 3. On the other hand, since $\lambda x \leq \frac{3}{8}\lambda^2 + \frac{2}{3}x^2$ for any λ and x , we have

$$\begin{aligned} \mathbb{E} \left[\exp \left(\frac{\lambda \Gamma_l}{c_l \sigma} \right) \middle| \theta_{[l-1]} \right] &\leq \exp \left(\frac{3\lambda^2}{8} \right) \mathbb{E} \left[\exp \left(\frac{2\Gamma_l^2}{3c_l^2 \sigma^2} \right) \middle| \theta_{[l-1]} \right] \\ &\leq \exp \left(\frac{3\lambda^2}{8} \right) \mathbb{E} \left[\exp \left(\frac{2\Delta_l^2}{3\sigma^2} \right) \middle| \theta_{[l-1]} \right] \\ &\leq \exp \left(\frac{3\lambda^2}{8} + \frac{2}{3} \right), \end{aligned} \quad (57)$$

for any λ , where the third inequality follows from the concavity of $f(x) = x^{\frac{2}{3}}$ and condition 3. Combining (56) with (57) yields, for any λ ,

$$\mathbb{E} \left[\exp \left(\frac{\lambda \Gamma_l}{c_l \sigma} \right) \middle| \theta_{[l-1]} \right] \leq \exp \left(\frac{3\lambda^2}{4} \right),$$

and thus

$$\mathbb{E} [\exp (\lambda \Gamma_l) | \theta_{[l-1]}] \leq \exp \left(\frac{3\lambda^2 c_l^2 \sigma^2}{4} \right).$$

□

Lemma 6. Let $\theta_1, \dots, \theta_k$ be a sequence of i.i.d. random variables, $\Phi_l = \Phi(\theta_{[l]})$ be deterministic functions of $\theta_{[l]} = (\theta_1, \dots, \theta_l)$, and $c_1, \dots, c_k$ be a sequence of positive numbers. If the inequality

$$\mathbb{E} \left[\exp \left(\frac{\Phi_l^2}{\sigma^2} \right) \middle| \theta_{[l-1]} \right] \leq \exp(1)$$

holds for each $l \leq k$, then for any $\Omega \geq 0$, we have

$$\mathbb{P} \left(\sum_{l=1}^k c_l \Phi_l^2 \geq (1 + \Omega) \sum_{l=1}^k c_l \sigma^2 \right) \leq \exp(-\Omega).$$

Proof. Since $f(x) = e^x$ is convex, we have

$$\mathbb{E} \left[\exp \left(\frac{\sum_{l=1}^k c_l \Phi_l^2}{\sum_{l=1}^k c_l \sigma^2} \right) \right] = \mathbb{E} \left[\exp \left(\frac{\sum_{l=1}^k c_l \sigma^2 \frac{\Phi_l^2}{\sigma^2}}{\sum_{l=1}^k c_l \sigma^2} \right) \right] \leq \frac{\sum_{l=1}^k c_l \sigma^2 \mathbb{E} \left[\exp \left(\frac{\Phi_l^2}{\sigma^2} \right) \right]}{\sum_{l=1}^k c_l \sigma^2}.$$

For each l , we have $\mathbb{E} \left[\exp \left(\frac{\Phi_l^2}{\sigma^2} \right) \right] = \mathbb{E} \left[\mathbb{E} \left[\exp \left(\frac{\Phi_l^2}{\sigma^2} \right) \middle| \theta_{[l-1]} \right] \right] \leq \exp(1)$, and thus

$$\mathbb{E} \left[\exp \left(\frac{\sum_{l=1}^k c_l \Phi_l^2}{\sum_{l=1}^k c_l \sigma^2} \right) \right] \leq \exp(1).$$

Then Markov inequality gives, for all $\delta > 0$,

$$\mathbb{P} \left(\exp \left(\frac{\sum_{l=1}^k c_l \Phi_l^2}{\sum_{l=1}^k c_l \sigma^2} \right) \geq \delta \right) \leq \frac{\exp(1)}{\delta}.$$

Therefore, for any $\Omega \geq 0$, by letting $\delta = \exp(1 + \Omega)$ and using the monotonicity of $f(x) = e^x$, we conclude that

$$\mathbb{P} \left(\sum_{l=1}^k c_l \Phi_l^2 \geq (1 + \Omega) \sum_{l=1}^k c_l \sigma^2 \right) \leq \frac{\exp(1)}{\exp(1 + \Omega)} = \exp(-\Omega).$$

Hence, we arrive at the desired concentration bound. □

D Proof of Remark 1

Here we show that the convergence rate in Theorem 5 can be improved to $\mathcal{O}\left(\frac{\log \frac{1}{\beta} \log^{\frac{1+\varepsilon}{2}} k}{\sqrt{k}}\right)$ by slightly modifying the stepsize.

Theorem 6. *Suppose that Assumptions 1, 2 and 3 hold. For any β , the sequence $\{x_k\}$ generated by SGDM (28) with stepsize $\eta_k = \frac{1}{16L^2 \log^{(1+\varepsilon)}(k+2)}$, where $0 < \varepsilon \leq 1$ and initialization $x_0 = x_1$ satisfies*

$$\mathbb{P}\left(\text{for any } k \geq 0, f(x_k) - f^* \leq \frac{\left(C_1 + C_2 \log \frac{1}{\beta}\right) \log^{\frac{1+\varepsilon}{2}}(k+2)}{\sqrt{k+1}}\right) \geq 1 - 2\beta,$$

where $C_1 = L\gamma_2\mathcal{E}(0) + L\sigma^2(1 + \sigma^2\gamma_1\gamma_2)\gamma_1$, $C_2 = L\gamma_2 + L\sigma^2(1 + \sigma^2\gamma_1\gamma_2)\gamma_1$, $\gamma_1 = \sum_{k=1}^{\infty} \frac{16\eta_k}{k}$, and $\gamma_2 = \prod_{k=1}^{\infty} (1 + \frac{16\eta_k}{k}\sigma^2)$.

Proof. As in Section 4, we let $a_k = \frac{16\eta_k}{k}$, $\gamma_1 = \sum_{k=1}^{\infty} a_k$, and $\gamma_2 = \prod_{k=1}^{\infty} (1 + a_k\sigma^2)$. It is straightforward to verify that these two constants are finite for $a_k = \frac{16\eta_k}{k} = \frac{1}{L^2 k \log^{(1+\varepsilon)}(k+2)}$.

The same argument as the proof of Theorem 5 yields that

$$\mathbb{P}\left(\sup_{k \geq 0} \mathcal{E}(k) \geq \gamma_2 \left(\mathcal{E}(0) + \log \frac{1}{\beta}\right) + \left(1 + \log \frac{1}{\beta}\right) \sigma^2(1 + \sigma^2\gamma_1\gamma_2)\gamma_1\right) \leq 2\beta.$$

Since $4\sqrt{(k+1)\eta_k}(f(x_k) - f^*) \leq \mathcal{E}(k)$, we conclude that

$$\mathbb{P}\left(\forall k \geq 0, f(x_k) - f^* \leq \frac{\left(C_1 + C_2 \log \frac{1}{\beta}\right) \log^{\frac{1+\varepsilon}{2}}(k+2)}{\sqrt{k+1}}\right) \geq 1 - 2\beta,$$

where $C_1 = L\gamma_2\mathcal{E}(0) + L\sigma^2(1 + \sigma^2\gamma_1\gamma_2)\gamma_1$, and $C_2 = L\gamma_2 + L\sigma^2(1 + \sigma^2\gamma_1\gamma_2)\gamma_1$. $\square$

E Additional Details and Results for the Experiments

In our artificial experiment, the objective function is $f(x) = \frac{1}{2}x^\top Ax$, and $A \in \mathbb{R}^{10 \times 10}$ is randomly generated. When the algorithm calls the gradient oracle at $x \in \mathbb{R}^{10}$, it outputs a stochastic gradient $Ax + \xi$, where $\xi \sim \mathcal{N}(0, 100 \cdot I_{10})$.

For the logistic regression task, the objective function is

$$f(\beta) = \frac{1}{N} \sum_{i=1}^N \left(y_i \log \frac{1}{1 + \exp(-x_i^\top \beta)} + (1 - y_i) \log \left(1 - \frac{1}{1 + \exp(-x_i^\top \beta)} \right) \right),$$

and we use `australian` dataset from LIBSVM library (Chang and Lin, 2011).

In the two image classification tasks, we set $\eta = 0.2$ for SGDM. The stepsize of SGD is tuned over $[10^{-4}, 1]$ by grid search (Bergstra and Bengio, 2012), and we report the results with the best choices, 0.005 for the MNIST task and 0.01 for the CIFAR-10 task. In the MNIST task, the model is trained for 40 epochs, while in the CIFAR-10 task, the model is trained for 400 epochs.

In Figures 3, 4, 5 and 6 we present another three individual runs of SGDM and SGD for the four experiments. All these results confirm that SGDM has faster convergence and smaller oscillations than SGD.

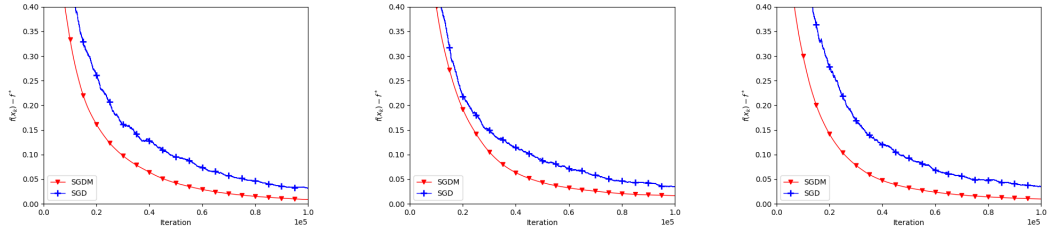


Figure 3: Three individual runs of SGDM and SGD on the artificial experiment.

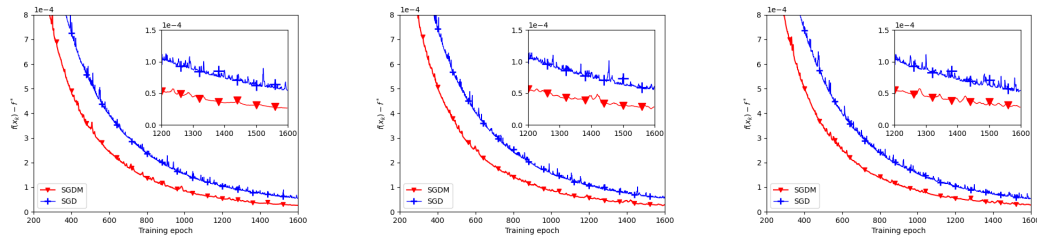


Figure 4: Three individual runs of SGDM and SGD on the logistic regression task. The subplots are zoomed in to provide better views of the last 1/4 training processes.

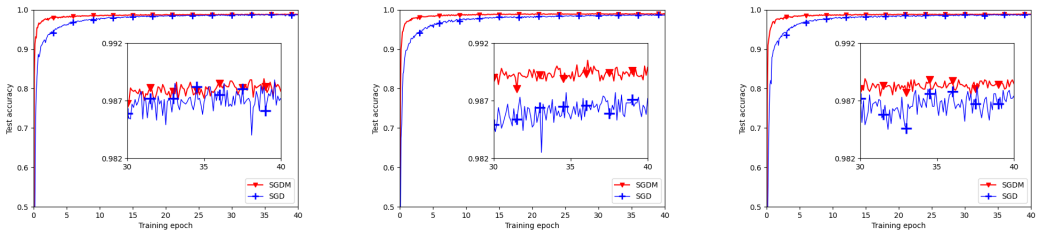


Figure 5: Three individual runs of SGDM and SGD on the MNIST task. The subplots are zoomed in to provide better views of the last 1/4 training processes.

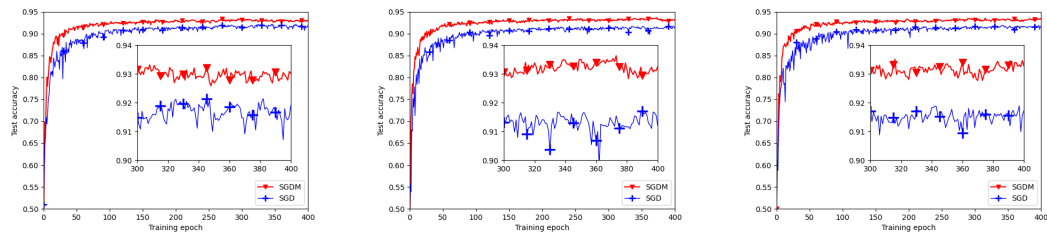


Figure 6: Three individual runs of SGDM and SGD on the CIFAR-10 task. The subplots are zoomed in to provide better views of the last 1/4 training processes.